

FYP PROJECT IDEAS - QUICK REFERENCE

IDEA 1: Cursor for Product Managers

Problem: Product managers waste 3-4 weeks manually analyzing customer interviews, finding patterns, writing PRDs, and creating Jira tickets. Insights get lost, requirements are unclear, and features get built wrong.

Solution: AI platform that automates product discovery. Upload interviews → AI transcribes, analyzes patterns, extracts insights → Ask "what should we build?" → AI generates complete PRD + broken-down Jira tickets → One-click export to Jira/Linear/Notion.

Tech Stack:

- Frontend: React.js, Redux Toolkit, Tailwind CSS, React Dropzone, Socket.io-client
- Backend: Node.js, Express.js, JWT authentication, Bull/Redis (background jobs)
- Database: MongoDB Atlas, Mongoose, GridFS for audio/video files
- AI: Gemini 2.0 API with multi-agent architecture (transcriber, analyst, writer, task decomposer)
- Integrations: Jira API, Linear API, Notion API, AWS S3 for file storage
- DevOps: Docker, GitHub Actions, Vercel (frontend), Railway (backend)

IDEA 2: AI Product Manager for Solo Founders

Problem: Solo founders have no product experience and can't afford PMs. They build wrong features, ignore customer feedback, waste months, and 90% fail because they built something nobody wants.

Solution: AI mentor that guides founders from idea to launch. Validates ideas with 20+ questions → Provides interview scripts → Simulates customer conversations for practice → Generates 6-month roadmap → Creates launch checklists → Acts as growth coach with weekly analytics insights.

Tech Stack:

- Frontend: React.js, Redux Toolkit, Tailwind CSS, Socket.io-client, Recharts
- Backend: Node.js, Express.js, JWT authentication, Socket.io (real-time chat)
- Database: MongoDB Atlas, Mongoose
- AI: Gemini 2.0 API with persona-based agents (validator, simulator, strategist, coach)
- Integrations: Stripe (payments), Mixpanel/PostHog (analytics), Resend (emails)
- DevOps: Docker, GitHub Actions, Vercel (frontend), Render (backend)